

Equivalence of Norms on Finite-Dimensional Vector Spaces

Definition. Let V be a finite-dimensional real or complex vector space with basis $e_1, \dots, e_n$. Writing $x \in V$ as $x = \sum_{i=1}^n x_i e_i$, the *supremum norm* with respect to this basis is defined by

$$\|x\|_\infty := \max_{1 \leq i \leq n} |x_i|.$$

Lemma. $\|\cdot\|_\infty$ is a norm on V .

Proof. Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. We verify the three norm axioms.

1. **Positive definiteness.** $\|x\|_\infty = \max_i |x_i| \geq 0$. If $\|x\|_\infty = 0$, then $|x_i| = 0$ for all i , hence $x = 0$. Conversely, $\|0\|_\infty = 0$.

2. **Absolute homogeneity.** For $\lambda \in \mathbb{F}$,

$$\|\lambda x\|_\infty = \max_i |\lambda x_i| = |\lambda| \max_i |x_i| = |\lambda| \|x\|_\infty.$$

3. **Triangle inequality.** For $x, y \in V$,

$$\|x + y\|_\infty = \max_i |x_i + y_i| \leq \max_i (|x_i| + |y_i|) \leq \max_i |x_i| + \max_i |y_i| = \|x\|_\infty + \|y\|_\infty. \quad \square$$

Definition. Two norms $\|\cdot\|$ and $\|\cdot\|'$ on a vector space are *equivalent* if there exist constants $c, C > 0$ such that

$$c\|x\| \leq \|x\|' \leq C\|x\| \quad \text{for all } x.$$

Remark. One can show that two norms are equivalent if and only if they generate the same topology. The term “equivalent” is thus justified.

Main Theorem

Theorem. All norms on a finite-dimensional real or complex vector space are equivalent.

Proof. Let V be a finite-dimensional real or complex vector space of dimension n with basis $e_1, \dots, e_n$, and let $\|\cdot\|$ be any norm on V . We show that $\|\cdot\|$ is equivalent to the supremum norm $\|\cdot\|_\infty$ defined with respect to this basis. Since equivalence is transitive and symmetric, this implies that any two norms on V are equivalent.

We establish the two required inequalities separately.

Step 1: Upper bound.

For any $x = \sum_{i=1}^n x_i e_i \in V$, the triangle inequality gives

$$\|x\| = \left\| \sum_{i=1}^n x_i e_i \right\| \leq \sum_{i=1}^n |x_i| \cdot \|e_i\| \leq \|x\|_\infty \sum_{i=1}^n \|e_i\|.$$

Setting $C := \sum_{i=1}^n \|e_i\| > 0$, we obtain

$$\|x\| \leq C \|x\|_\infty \quad \text{for all } x \in V. \quad (1)$$

Step 2: Continuity of $\|\cdot\|$ with respect to $\|\cdot\|_\infty$.

Let $(x^{(k)})$ be a sequence with $\|x^{(k)} - x\|_\infty \rightarrow 0$. Then by (1),

$$|\|x^{(k)}\| - \|x\|| \leq \|x^{(k)} - x\| \leq C \|x^{(k)} - x\|_\infty \rightarrow 0,$$

where the first inequality is the reverse triangle inequality. Hence $\|\cdot\|$ is continuous with respect to $\|\cdot\|_\infty$.

Step 3: The unit sphere is sequentially compact.

Let $S := \{x \in V : \|x\|_\infty = 1\}$. We claim S is sequentially compact.

Let $(x^{(k)})$ be any sequence in S . Writing $x^{(k)} = \sum_{i=1}^n x_i^{(k)} e_i$, each coordinate sequence $(x_i^{(k)})_{k \geq 1}$ is bounded (since $|x_i^{(k)}| \leq \|x^{(k)}\|_\infty = 1$). By the Bolzano–Weierstrass theorem, applied iteratively to each of the n coordinates, we extract a subsequence along which all n coordinate sequences converge. The limit point $x^* = \sum_{i=1}^n x_i^* e_i$ satisfies $\|x^*\|_\infty = 1$ (since the sup norm is continuous and each $\|x^{(k)}\|_\infty = 1$), so $x^* \in S$. Thus S is sequentially compact.

Step 4: Lower bound.

Since $\|\cdot\|$ is continuous with respect to $\|\cdot\|_\infty$ (Step 2) and S is sequentially compact (Step 3), $\|\cdot\|$ attains its minimum on S . Let

$$c := \min_{x \in S} \|x\|.$$

We claim $c > 0$. Indeed, if $c = 0$, then there exists $x \in S$ with $\|x\| = 0$, implying $x = 0$. But $x \in S$ means $\|x\|_\infty = 1 \neq 0$, a contradiction. Hence $c > 0$.

Now let $x \in V$ be nonzero. Setting $y := x/\|x\|_\infty \in S$, we have $\|y\| \geq c$, i.e.,

$$\frac{\|x\|}{\|x\|_\infty} \geq c,$$

which gives

$$c \|x\|_\infty \leq \|x\|. \quad (2)$$

(The inequality holds trivially for $x = 0$.)

Conclusion.

Combining (1) and (2),

$$c \|x\|_\infty \leq \|x\| \leq C \|x\|_\infty \quad \text{for all } x \in V,$$

with $0 < c \leq C$. Hence $\|\cdot\|$ and $\|\cdot\|_\infty$ are equivalent. $\square$

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